**Foundations of Data Engineering and Analytics**

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# Sheet 3: Web Crawling and Univariate EDA

You are investigating what an online job platform appears to tell us about European technology careers. Collect and inspect data from DevHire Europe (https://devhire-europe.pages.dev), a fictional educational website built by the University of Innsbruck using public data. All advertisements and companies are fictional. We will also use this dataset in Sheet 4.

## Before you start: understand the website and plan your crawler

Make yourself familiar with the following concepts and tools *before* starting your crawler. Choose an approach suited to the website and be prepared to justify your decisions. The tools listed below are suggestions.

- Client-side rendering, JavaScript, and the DOM.
- Headless browser crawling, browser automation (such as Playwright, Selenium, etc.).
- `robots.txt`: `User-agent`, `Allow`, `Disallow`, `Sitemap`, and `Crawl-delay`.
- HTTP headers and `User-Agent`.
- Rate limiting, concurrency, shared IP addresses, HTTP status codes, and `Retry-After`.

**Plan ahead and crawl responsibly.** Inspect the platform's `robots.txt` and data-use information. Choose your request rate, concurrency, and error-handling strategy carefully to avoid unnecessary load and reduce the risk of being blocked. Start early, test your approach on a small scale, and allow time for adjustments and recovery.
Being blocked as a result of your crawling strategy does not exempt you from completing the exercise and is not, by itself, grounds for an extension. Respect access restrictions, document any difficulties, and contact course staff in good time if problems arise.

## Exercise 1 (Responsible Web Crawling) [5 Points]

a) *1.5 Points* **Plan and test your approach.** Before collecting the full dataset, briefly document your approach in your notebook:
   - Inspect the website and justify your choice of tools and collection strategy.
   - Explain how you will collect the data responsibly and handle interruptions or unexpected responses.
   - Test your approach on a small sample and discuss what you learned before scaling up.

b) *3 Points* **Collect the 2026 snapshot.** Implement a crawler for DevHire Europe's job listings, using tools of your choice. Collect and store the information presented on the listing and advertisement pages. The platform contains 1,200 advertisements in this snapshot. Your solution should handle the website's navigation appropriately.

   Collect the following information:

- Identifiers and description: `job_id`, `survey_year`, `title`, `role_family`, `seniority`.
- Company and location: `company_name`, `company_size`, `industry`, `country`, `city`, `work_mode`.
- Advertised requirements and pay: `required_experience_years`, `technologies`, `salary_min_eur`, `salary_max_eur`, `salary_disclosed`.
- Publication date: `posted_at`.

Develop a suitable strategy for storing the collected data and making it usable for analysis. Consider how to represent technology tags and other multi-valued fields, handle missing information and collection failures, and avoid duplicates. Explain your choices and how you would recover from an interrupted crawl.

c) *0.5 Points* **Validate your collection.** Report the row count, coverage against the platform's total, duplicate IDs, data types, and missing-value rates. Check salary-bound ordering and whether undisclosed salaries are represented consistently. Document any extraction errors, missing fields, exclusions, or unresolved failures.

## Exercise 2 (Univariate Exploratory Analysis) [5 Points]

Analysing and summarizing data requires a solid understanding of the dataset. Investigate individual variables to identify patterns, spot anomalies, and verify assumptions. Use the job advertisements collected in Exercise 1 and apply appropriate data cleaning and handling strategies. The main goal is to gain a thorough understanding of the variables in your dataset. Base your interpretations on your calculations and charts, and report the number of observations used in each analysis.
For the salary analyses, use the *salary midpoint*, the mean of the advertised minimum and maximum. Compute it only when both bounds are available and report how many advertisements remain.

a) *1 Point* Use **tabulation** to compute the absolute and relative frequency of advertisements per *country*. Interpret your findings.

b) *3 Points* For the following tasks, focus on the *salary midpoint*:

- Use measures of location to describe its central tendency by calculating the mean, 20% trimmed mean, median, and mode. Interpret your results and explain the differences between these measures. Discuss whether the mode is informative for these salary values. Visualize the distribution using a histogram, a density plot, and a boxplot combined with a density plot.
- Identify its spread using the variance, standard deviation, and interquartile range.
- Identify and explain whether the salary values are skewed. How does their tail behaviour compare with a normal distribution? Use graphical and non-graphical methods to support your arguments. Identify any outliers and discuss their effect on your summaries.

c) *1 Point* Investigate whether any numerical variable in the dataset is (nearly) normally distributed. Support your conclusion with appropriate analyses. What does your finding imply, and why can it be useful to know whether values are normally distributed?

**Submission.** Submit the reproducible notebook (`.ipynb`), any separate crawler source, your collected dataset, dependency instructions, and an exported `.html` or `.pdf` notebook to OLAT. Include your collection strategy, small-scale test results, calculations, charts, and interpretations.
**Important:** Submit your solution to OLAT and mark your solved exercises with the provided checkboxes. The deadline ends at 23:59 on the day before the discussion.